Building Materials

Wei, Ya （魏亚）
Department of Civil Engineering

Tel. : +86-10-62771646
Email: yawei@tsinghua.edu.cn

Chapter One Introduction
Chapter Two The Fundamentals
Chapter Three Metal
Chapter Four Concrete
Chapter Five Asphalt
Chapter Six Masonry
Chapter Seven Wood

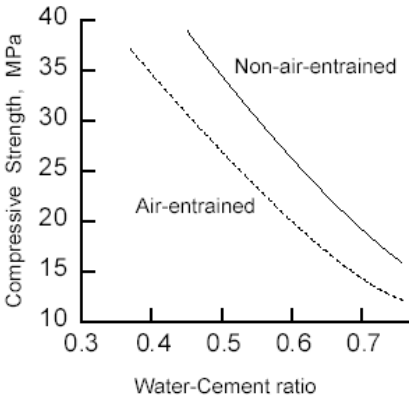
Chapter Four：Concrete

1. Introduction
2. Compositions
3. Structure
4. **Strength and Failure**
5. **Mix Proportioning**
6. Early-age Properties
7. Volume Changes
8. Durability



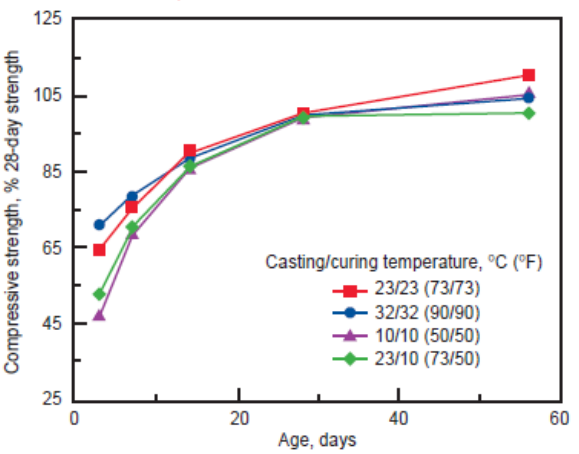
4.4 Strength and Failure >> compressive strength

Effect of Air Entraining



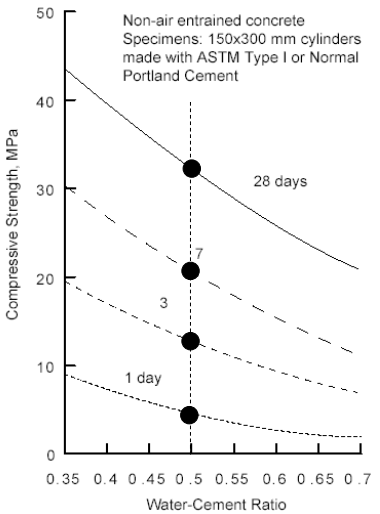
4.4 Strength and Failure >> compressive strength

Effect of Curing Temperature



Crossover effect

4.4 Strength and Failure >> Compressive strength



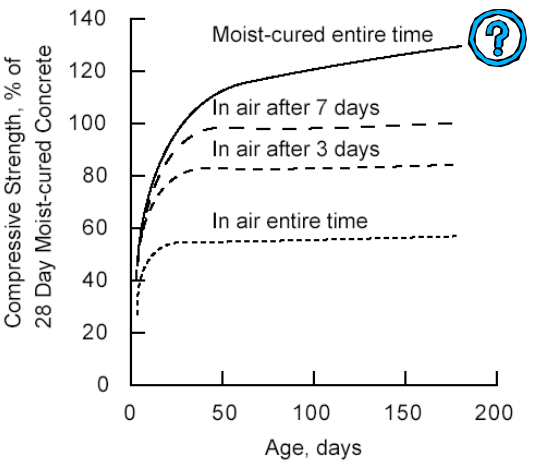
ACI 209 Committee

$$f_{cm}(t) = f_{c28}\left(\frac{t}{4+0.85t}\right)$$

f_{cm}: compressive strength
f_{c28}: 28day's strength
t: time

4.4 Strength and Failure >> Compressive strength

Effect of Moisture Curing





4.4 Strength and Failure >> compressive strength

Measures to control concrete temperature



Liquid nitrogen added directly into a truck mixer at the ready mix plant is an effective method of reducing concrete temperature for mass concrete placements or during hot-weather concreting.



Hoses laying on soil to defrost subgrade

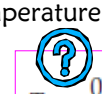


warming the forms while fresh concrete is pumped in

4.4 Strength and Failure >> compressive strength

Calculating mixture temperature

The approximate temperature of concrete can be calculated from the temperatures of its ingredients by using the following equation:



$$T = \frac{0.22(T_a M_a + T_c M_c) + T_w M_w + T_{wa} M_{wa}}{0.22(M_a + M_c) + M_w + M_{wa}}$$

T = temperature of the freshly mixed concrete, ° C

Ta , Tc, Tw, and Twa = temperature in ° C of aggregates, cement, added mixing water, and free water on aggregates, respectively.

Ma , Mc, Mw, and Mwa = mass, kg (lb), of aggregates, cementing materials, added mixing water, and free water on aggregates, respectively

4.4 Strength and Failure >> compressive strength

Calculating mixture temperature

Example calculations for initial concrete temperature

Material	Mass, *M*, kg	Specific heat kJ/kg • K	Joules to vary temperature, 1°C	Initial temperature of material, *T*, °C	Total joules in material
	(1)	(2)	(3) Col.1 x Col. 2	(4)	(5) Col. 3 x Col. 4
Cement	335 (M_c)	0.92	308	66 (T_c)	20,328
Water	123 (M_w)	4.184	515	27 (T_w)	13,905
Total aggregate	1839 (M_a)	0.92	1692	27 (T_a)	45,684
			2515		79,917

Initial concrete temperature = 79,917/2515 = 31.8° C
To achieve 1° C reduction in initial concrete temperature:
Cement temperature must be lowered =2515/308 = 8.2° C
Or water temperature dropped =2515/515 = 4.9° C
Or aggregate temperature cooled =2515/1692 = 1.5° C

4.4 Strength and Failure >> strength tests

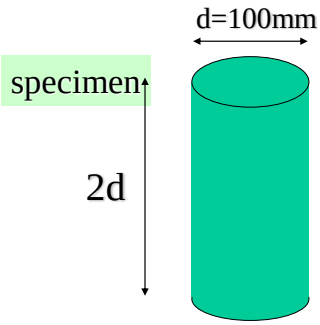
Types of concrete tests

- Destructive Test
  - a) Compression strength test
    - Cube test
    - Cylinder test
  - b) Tensile strength test
    - Direct Tension Test
    - Split tensile Test
    - Flexural Test
- Non Destructive Test

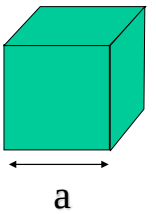


4.4 Strength and Failure >> compressive strength

Testing Method



China：a = 150 mm



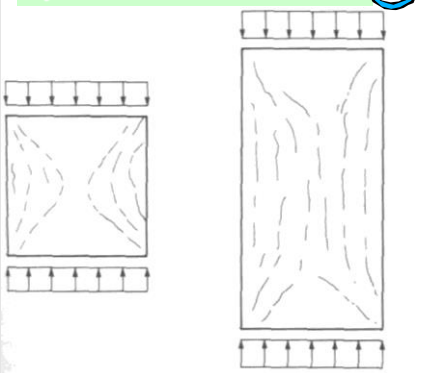
cylinder	cube
（USA、France、Japan）	（UK、Germany、China）

4.4 Strength and Failure >> compressive strength

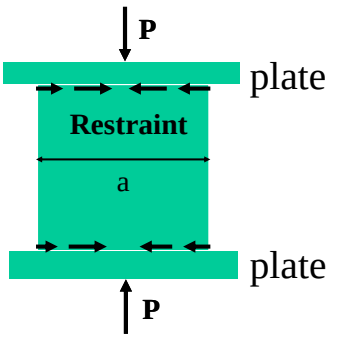
Testing Method

$f_{cylinder} \times 1.25 = f_{cube}$

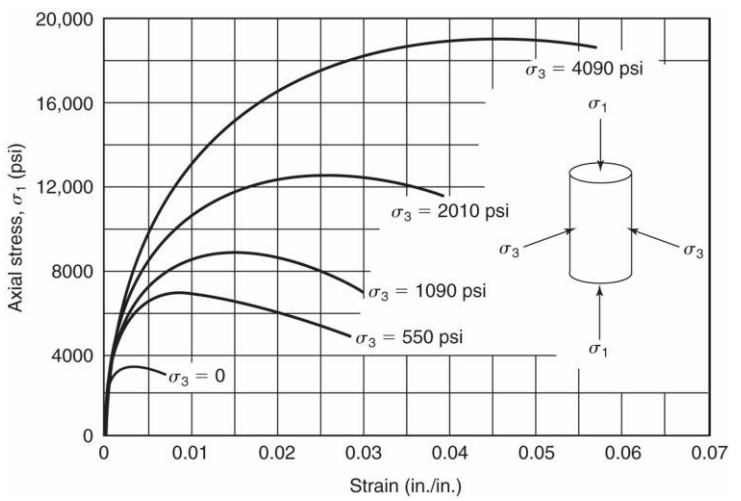
Compressive strength:
cylinder = cubic ?



Restraint from plate



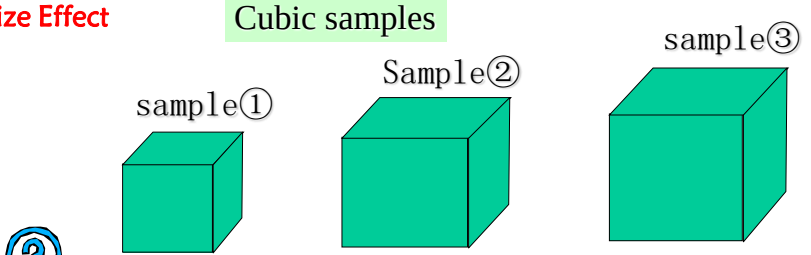
Concrete Confinement



4.4 Strength and Failure >> compressive strength

Size Effect

Cubic samples



Size effect: strength① > strength② > strength③

a (mm)	100	150 (standard)	300	450
Strength correction factor	**0.95**	**1.00**	**1.15**	**1.36**

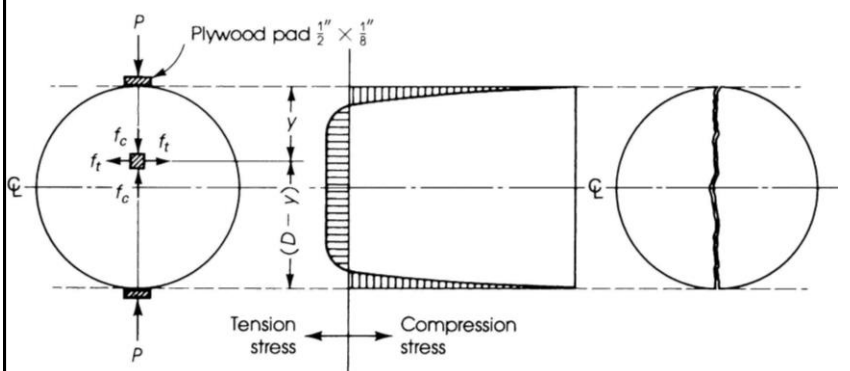
4.4 Strength and Failure >> Tensile strength

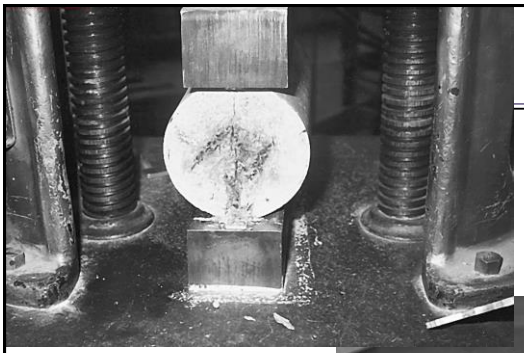
Split tensile tests

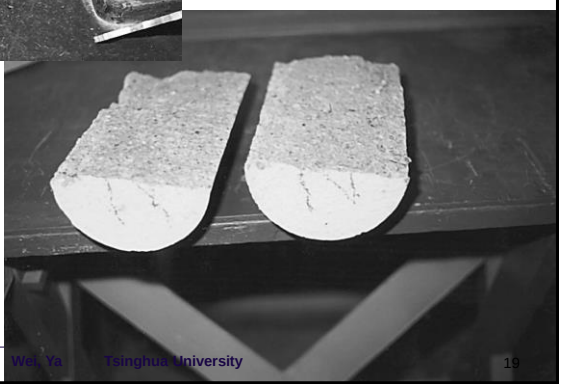
Split tensile: $f_s = \dfrac{2P}{\pi l d}$

Diameter d, height l

Greater than direct tensile by 10%-15%



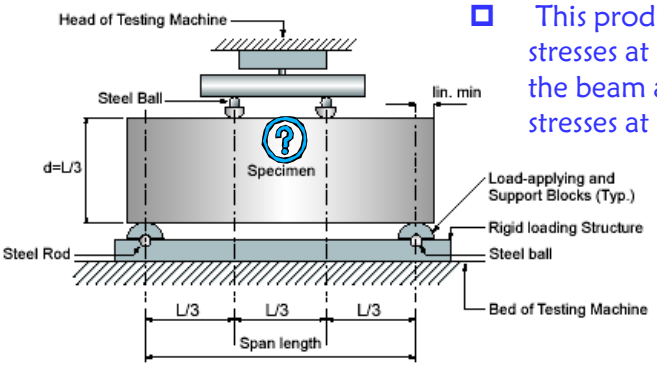




4.4 Strength and Failure >> strength tests

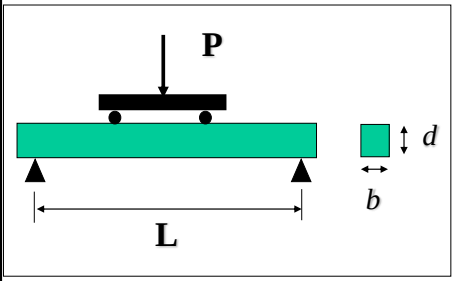
Flexural strength tests

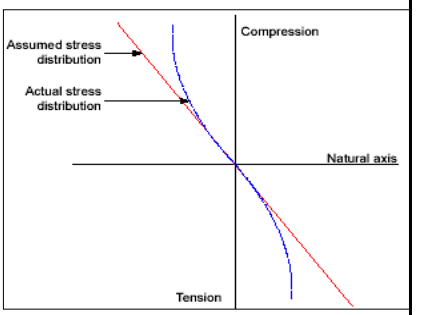
- Most common method for measuring the tensile strength of concrete.
- This produces tensile stresses at the bottom of the beam and compressive stresses at the top.



4.4 Strength and Failure >> strength tests

Flexural strength tests





Flexural strength $f_b = \dfrac{PL}{bd^2}$ Greater than direct tensile by 50%-100%

4.4 Strength and Failure >> compressive strength

Flexural Strength



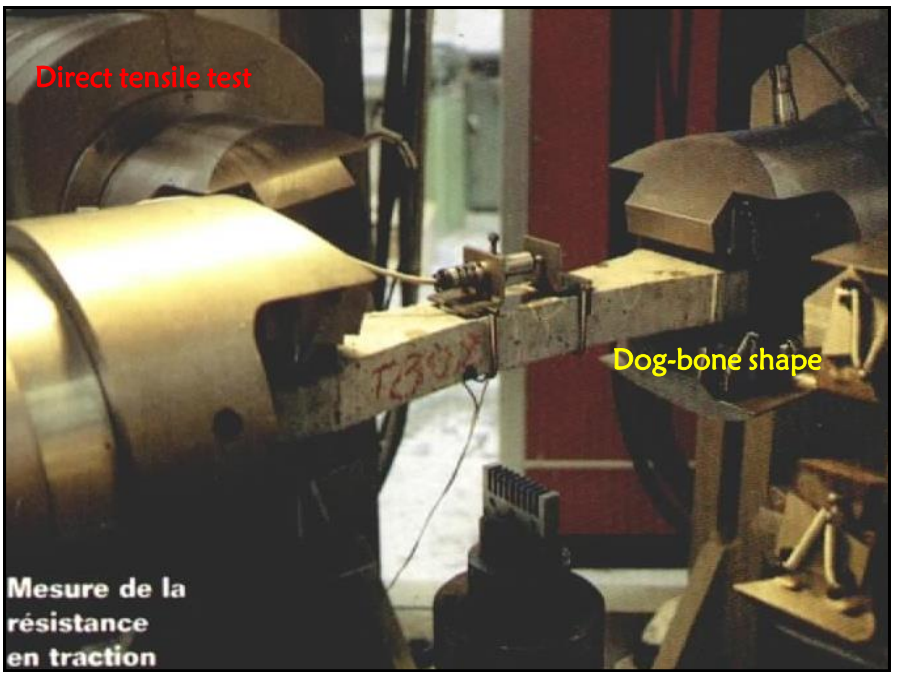
ACI18.3.3

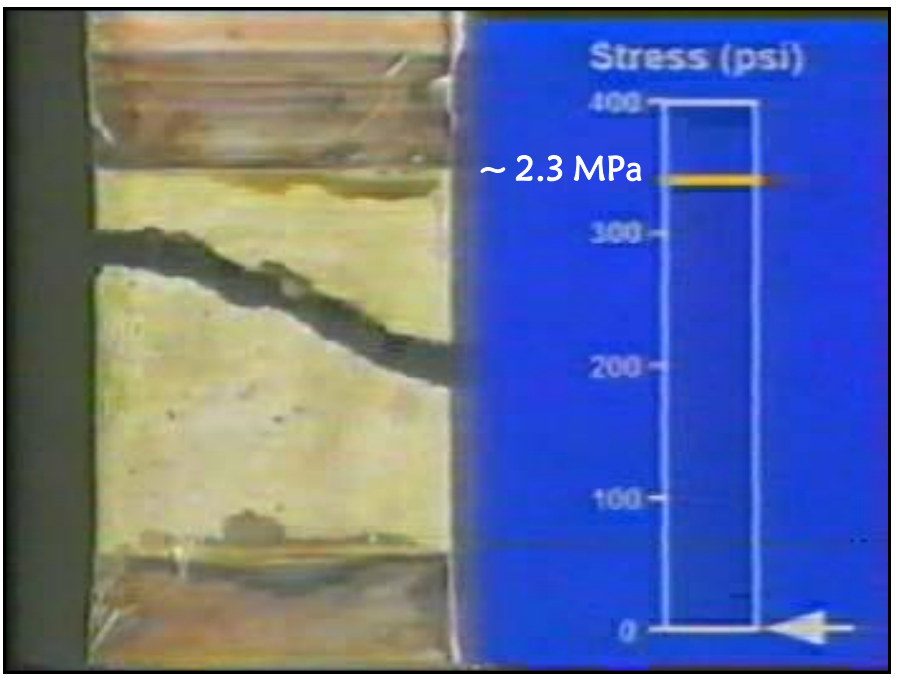
$f_r = 7.5\lambda\sqrt{f_c'}$ (psi)

$\lambda = 1.0$ for normal weight concrete

$\lambda = 0.85$ for sand lightweigh t concrete

$\lambda = 0.75$ for all lightweigt concrete


Direct tensile test
Dog-bone shape
Mesure de la
résistance
en traction


Stress (psi)
400
~ 2.3 MPa
300
200
100
0

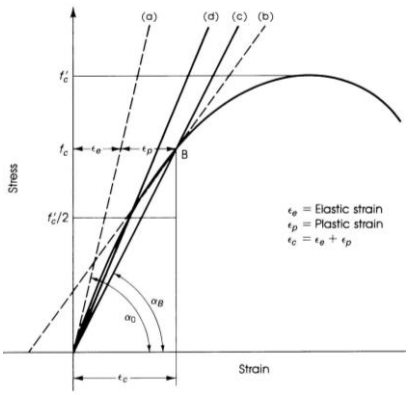
4.4 Strength and Failure >> strength tests

Modulus of Elasticity



ACI 8.5.1

$$E_c = 57{,}000\sqrt{f_c'} \quad \text{(psi)}$$



4.4 Strength and Failure >> strength tests

Nondestructive Test: Rebound Hammer Test



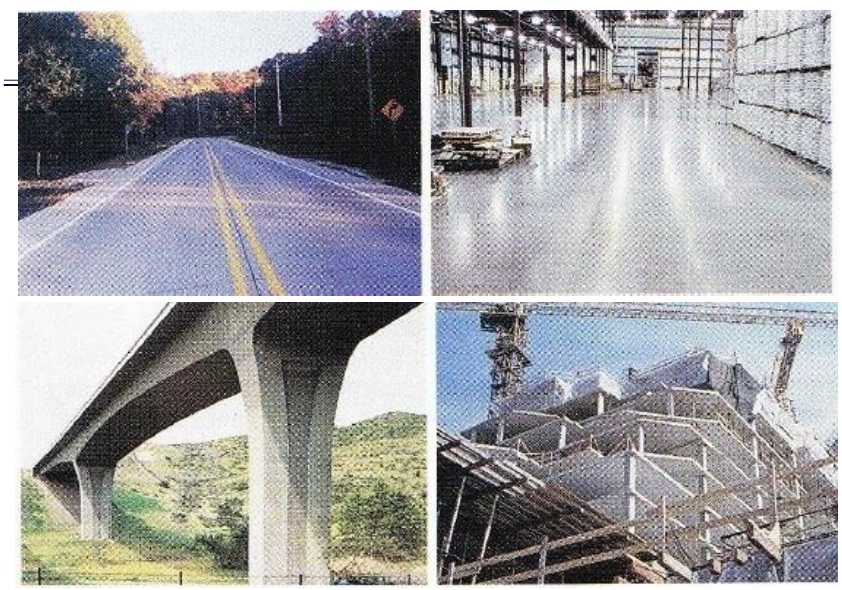
What is the defect of this method?
What would the contractor probably do?





Chapter Four：Concrete

1. Introduction
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5. **Mix Proportioning**
6. Fresh and Early-age Properties
7. Volume Changes
8. Durability





1:2:3 ??





4.5 Mix Proportioning

In the design of concrete mixes, three principal requirements for concrete are of importance:

- Quality
- Workability
- Economical

4.5 Mix Proportioning

- *Quality* of concrete is measured by its strength and durability. The principal factors affecting the strength of concrete are sound aggregates, w/c ratio, and the extent to which hydration has progressed. Durability of concrete is the ability of the concrete to resist disintegration due to freezing and thawing and chemical attack.

- *Workability*

The ease with which a concrete mixture can be mixed, transported, placed in the form and compacted without any segregation.

- *Economical*

the cost of producing good quality concrete is an important consideration in the overall cost of the construction project.

4.5 Mix Proportioning

Mixture proportioning:
refers to the process of determining the quantities of concrete ingredients, using local materials, to achieve the specified characteristics of the concrete.

A properly proportioned concrete mix should possess these qualities:

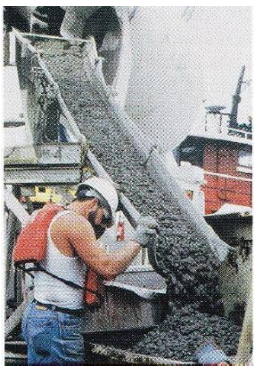
1. Durability, strength, and uniform appearance of the hardened concrete
2. Acceptable workability of the freshly mixed concrete
3. Economy

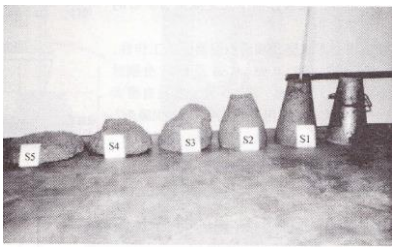




Trial batching verifies that a concrete mixture meets design requirements prior to use in construction.

4.5 Mix Proportioning





4.5 Mix Proportioning

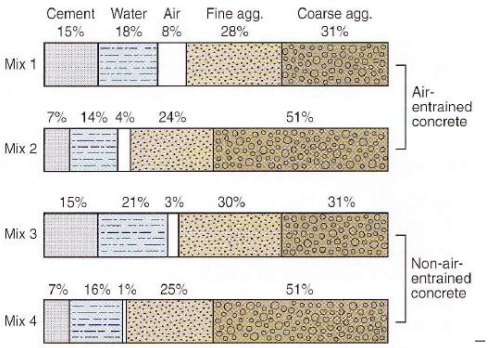
The influence of ingredients on properties of concrete

Ingredient	Quality	Workability	Economy
Aggregate	Increases	Decreases	Increases
Portland Cement	Increases	Increases	Decreases
Water	Decreases	Increases	Increases

4.5 Mix Proportioning



Volume or mass-based proportioning?



4.5 Mix Proportioning

Selecting mixture characteristics:

- Before a concrete mixture can be proportioned, mixture characteristics are selected based on the intended use of the concrete, the exposure conditions, the size and shape of building elements, and the physical properties of the concrete (such as frost resistance and strength) required for the structure.

- Since most of the desirable properties of hardened concrete depend primarily upon the quality of the cementitious paste, the first step in proportioning a concrete mixture is the selection of the appropriate water-cementing materials ratio for the durability and strength needed.

4.5 Mix Proportioning

Overall procedure



Stage 1 – determine the free-water/cement ratio for the targeted mean strength

Stage 2 – determine the free-water content for the targeted workability

Stage 3 – combines the results of stages 1 and 2 to give the cement content

Stage 4 – deals with the determination of the total aggregate content

Stage 5 – deals with the selection of the fine and coarse aggregate content

4.5 Mix Proportioning

Step 1: Choice of slump

Step 2: Choice of maximum size of aggregate

Step 3: Selection of water/cement ratio

Step 4: Estimation of mixing water and air content

Step 5: Calculation of cement content and type

Step 6: Estimation of coarse aggregate content

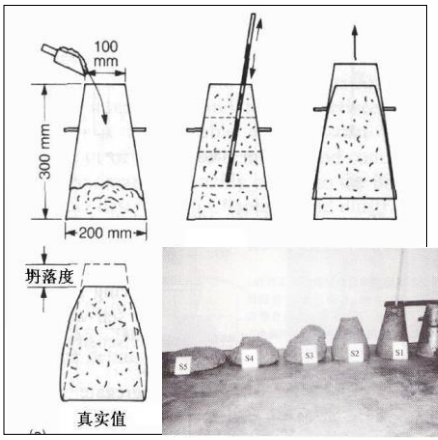
Step 7: Estimation of fine aggregate content

Step 8: Adjustment of aggregate moisture

Step 9: Trial batch adjustments

4.5 Mix Proportioning

Step 1: Choice of slump





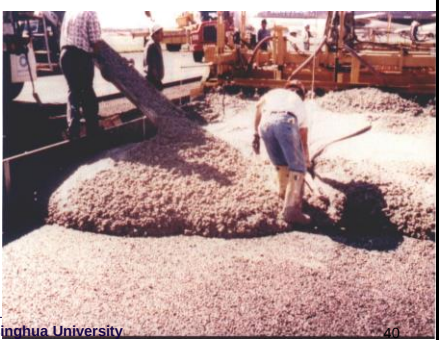


4.5 Mix Proportioning

Step 1: Choice of slump

	slump (mm)
Roller-compacted concrete	0
Slipform paving concrete	30 ~ 50





4.5 Mix Proportioning

Step 1: Choice of slump

	slump (mm)
Pumping concrete	100 ~ 200
Self-compacting concrete	> 240





4.5 Mix Proportioning

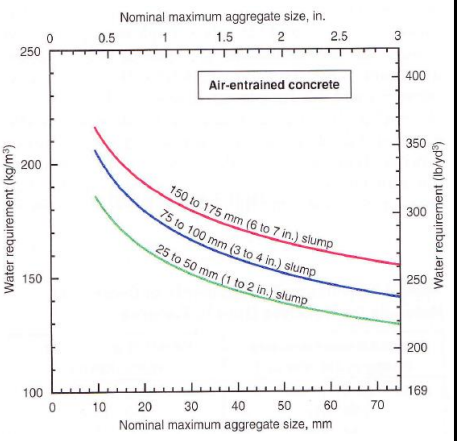
Step 1: Choice of slump

Types of Construction	Slump (in.)	
	Maximum*	Minimum
Reinforced foundation walls and footings	3	1
Plain footings, caissons, and substructure walls	3	1
Beams and reinforced walls	4	1
Building columns	4	1
Pavements and slabs	3	1
Mass concrete	2	1

4.5 Mix Proportioning

Step 2: Choice of maximum size of aggregate

- The amount of mixing water required to produce a unit volume of concrete of a given slump is dependent on the shape and the maximum size and amount of coarse aggregate.
- Larger sizes minimize the water requirement and thus allow the cement content to be reduced.
- Rounded aggregate requires less mixing water than a crushed aggregate in concretes of equal slump.



4.5 Mix Proportioning

Step 2: Choice of maximum size of aggregate

- Maximum size of aggregate should be the largest available so long it is consistent with the dimensions of the structure.
- In practice the dimensions of the forms or the spacing of the rebars controls the maximum CA size.
- The maximum CA size should not exceed:

1) one-fifth of the narrowest dimension between sides of forms,
2) one-third the depth of slabs,
3) 3/4-ths of the minimum clear spacing between individual reinforcing bars, bundles of bars, or pre-tensioning strands.

Special Note: When high strength concrete is desired, best results may be obtained with reduced maximum sizes of aggregate since these produce higher strengths at a given w/c ratio.

4.5 Mix Proportioning

Step 3: Selection of water/cement ratio

- The water-cementitious material ratio is simply the mass of water divided by the mass of cementitious material (portland cement, blended cement, fly ash, slag, silica fume, and natural pozzolans).
- The required water/cement ratio is determined by strength, durability and finishability.

4.5 Mix Proportioning

Step 3: Selection of water/cement ratio

- The water-cementitious material ratio selected for mix design must be the lowest value required to meet anticipated exposure conditions.

Maximum Water-Cementitious Material Ratios and Minimum Design Strengths for Various Exposure Conditions

Exposure condition	Maximum water-cementitious material ratio by mass for concrete	Minimum design compressive strength, f'_c, MPa (psi)
Concrete protected from exposure to freezing and thawing, application of deicing chemicals, or aggressive substances	Select water-cementitious material ratio on basis of strength, workability, and finishing needs	Select strength based on structural requirements
Concrete intended to have low permeability when exposed to water	0.50	28 (4000)
Concrete exposed to freezing and thawing in a moist condition or deicers	0.45	31 (4500)
For corrosion protection for reinforced concrete exposed to chlorides from deicing salts, salt water, brackish water, seawater, or spray from these sources	0.40	35 (5000)

4.5 Mix Proportioning

Step 3: Selection of water/cement ratio

When durability does not control, the water-cementitious materials ratio should be selected on the basis of concrete compressive strength.

Compressive strength at 28 days, MPa	Water-cementitious materials ratio by mass	
	Non-air-entrained concrete	Air-entrained concrete
45	0.38	0.30
40	0.42	0.34
35	0.47	0.39
30	0.54	0.45
25	0.61	0.52
20	0.69	0.60
15	0.79	0.70



4.5 Mix Proportioning

Step 3: Selection of water/cement ratio

As a result of the inherent variability of concrete in production, it is necessary to design a mix to have a mean strength greater than the specified characteristic strength. Thus:

Statistical constant

$$f_{cu,0} \geq f_{cu,k} + 1.645\sigma$$

for 10% defectives = 1.28
for 5% defectives = 1.645
for 2.5% defectives = 1.96
for 1% defectives = 2.33

Where,

$f_{cu,0}$: the target mean strength

$f_{cu,k}$: the specified characteristic strength

σ : the standard deviation

Concrete strength	C10 ~ C20	C25 ~ C40	C45 ~ C60
σ	4.0 MPa	5.0 MPa	6.0 MPa

4.5 Mix Proportioning

Step 4: Estimation of mixing water and air content

- In general the quantity of water per unit volume of concrete required to produce a given slump is dependent on the maximum coarse aggregate size, the shape and grading of both coarse aggregate and fine aggregate, as well as the amount of entrained air.

4.5 Mix Proportioning

Step 4: Estimation of mixing water and air content

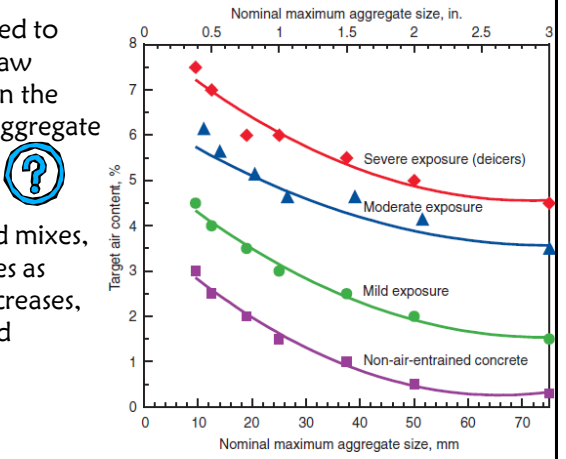
Approximate Mixing Water and Target Air Content Requirements for Different Slumps and Nominal Maximum Sizes of Aggregate (kg/m^3)

Slump, mm	Water, kilograms per cubic meter of concrete, for indicated sizes of aggregate*						
	9.5 mm	12.5 mm	19 mm	25 mm	37.5 mm	50 mm**	75 mm**
	Non-air-entrained concrete						
25 to 50	207	199	190	179	166	154	130
75 to 100	228	216	205	193	181	169	145
150 to 175	243	228	216	202	190	178	160
Approximate amount of entrapped air in non-air-entrained concrete, percent	3	2.5	2	1.5	1	0.5	0.3
	Air-entrained concrete						
25 to 50	181	175	168	160	150	142	122
75 to 100	202	193	184	175	165	157	133
150 to 175	216	205	197	184	174	166	154
Recommended average total air content, percent, for level of exposure:†							
Mild exposure	4.5	4.0	3.5	3.0	2.5	2.0	1.5
Moderate exposure	6.0	5.5	5.0	4.5	4.5	4.0	3.5
Severe exposure	7.5	7.0	6.0	6.0	5.5	5.0	4.5

4.5 Mix Proportioning

Step 4: Estimation of mixing water and air content

❑ The amount of air required to provide adequate freeze-thaw resistance is dependent upon the nominal maximum size of aggregate and the level of exposure.



❑ In properly proportioned mixes, the mortar content decreases as maximum aggregate size increases, thus decreasing the required concrete air content.

4.5 Mix Proportioning

Step 4: Estimation of mixing water and air content ACI

Mild Exposure: This exposure includes indoor or outdoor service in a climate where concrete will not be exposed to freezing or deicing agents.

Moderate Exposure: Service in a climate where freezing is expected but where the concrete will not be continually exposed to moisture or free water for long periods prior to freezing and will not be exposed to deicing or other aggressive chemicals. Examples include exterior beams, columns, walls, girders, or slabs that are not in contact with wet soil and are so located that they will not receive direct applications of deicing chemicals.

Severe Exposure: Concrete that is exposed to deicing or other aggressive chemicals or where the concrete may become highly saturated by continual contact with moisture or free water prior to freezing. Examples include pavements, bridge decks, curbs, gutters, sidewalks, canal linings, or exterior water tanks.

4.5 Mix Proportioning

Step 5: Calculation of cement content and type

- The cementing materials content is usually determined from the selected water-cementing materials ratio and water content.
- Minimum cement content requirements serve to ensure satisfactory durability and finishability, to improve wear resistance of slabs, and to guarantee a suitable appearance of vertical surfaces.

4.5 Mix Proportioning

Step 5: Calculation of cement content and type

Nominal maximum size of aggregate, mm (in.)	Cementing materials, kg/m³ (lb/yd³)*
37.5 (1½)	280 (470)
25 (1)	310 (520)
19 (¾)	320 (540)
12.5 (½)	350 (590)
9.5 (⅜)	360 (610)

4.5 Mix Proportioning

Step 5: Calculation of cement content and type

- For severe freeze-thaw, deicer, and sulfate exposures, it is desirable to specify: (1) a minimum cementing materials content of 335 kg per cubic meter of concrete, and (2) only enough mixing water to achieve the desired consistency without exceeding the maximum water-cementing materials ratios.

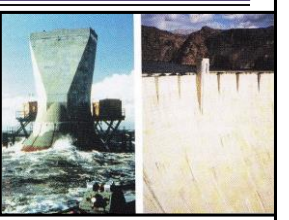
- Supplementary cementitious materials have varied effects on water demand and air contents. The addition of fly ash will generally reduce water demand and decrease the air content. Silica fume increases water demand and decreases air content. Slag and metakaolin have a minimal effect at normal dosages.

4.5 Mix Proportioning

Low heat cement

Step 5: Calculation of cement content and type





Early age strength cement



混凝土强度等级	≤C10	C15 ~ C25	C30 ~ C40	≥C45
水泥强度等级	32.5	32.5、42.5	42.5、52.5	52.5、62.5

4.5 Mix Proportioning

Example:

Conditions and Specifications:

Concrete is required for a pavement that will be exposed to moisture in a severe freeze-thaw environment. A specified compressive strength of 35 MPa is required at 28 days. Air entrainment is required. Slump should be between 25 mm and 75 mm. A nominal maximum size aggregate of 25 mm is required.

The materials available are:

Cement: with a relative density of 3.0

Coarse aggregate: Well-graded, 25-mm nominal maximum-size rounded gravel with an ovendry relative density of 2.68, absorption of 0.5% (moisture content at SSD condition) and ovendry rodded bulk density (unit weight) of 1600 kg/m^3. The laboratory sample for trial batching has a moisture content of 2%.

4.5 Mix Proportioning

Example:

Conditions and Specifications.

Fine aggregate: Natural sand with an ovendry relative density of 2.64 and absorption of 0.7%. The laboratory sample moisture content is 6%. The fineness modulus is 2.80.

Air-entraining admixture: Wood-resin type.

Water reducer: This particular admixture is known to reduce water demand by 10% when used at a dosage rate of 3 g (or 3 mL) per kg of cement. Assume that the chemical admixtures have a density close to that of water.

From this information, the task is to proportion a trial mixture that will meet the above conditions and specifications

4.5 Mix Proportioning

Overall procedure

Stage 1 – determine the free-water/cement ratio for the targeted mean strength

Stage 2 – determine the free-water content for the targeted workability

Stage 3 – combines the results of stages 1 and 2 to give the cement content

Stage 4 – deals with the determination of the total aggregate content

Stage 5 – deals with the selection of the fine and coarse aggregate content

4.5 Mix Proportioning

Example: (absolute volume method)

Strength. The design strength of 35 MPa is greater than the 31 MPa required in Table for the exposure condition.

Exposure condition	Maximum water-cementitious material ratio by mass for concrete	Minimum design compressive strength, f'_c, MPa (psi)
Concrete protected from exposure to freezing and thawing, application of deicing chemicals, or aggressive substances	Select water-cementitious material ratio on basis of strength, workability, and finishing needs	Select strength based on structural requirements
Concrete intended to have low permeability when exposed to water	0.50	28 (4000)
Concrete exposed to freezing and thawing in a moist condition or deicers	0.45	31 (4500)
For corrosion protection for reinforced concrete exposed to chlorides from deicing salts, salt water, brackish water, seawater, or spray from these sources	0.40	35 (5000)

4.5 Mix Proportioning

Example: (absolute volume method)

Strength.

Since no statistical data is available, the required compressive strength for proportioning from Table is: 35 + 8.5 = 43.5 MPa.

Required Average Compressive Strength When Data Are Not Available to Establish a Standard Deviation

Specified compressive strength, f'_c, MPa	Required average compressive strength, f'_{cr}, MPa
Less than 21	f'_c + 7.0
21 to 35	f'_c + 8.5
Over 35	1.10f'_c + 5.0

4.5 Mix Proportioning

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As a result of the inherent variability of concrete in production, it is necessary to design a mix to have a mean strength greater than the specified characteristic strength by a statistical amount termed as the margin. Thus:

Statistical constant

$$f_{cu,0} \geq f_{cu,k} + 1.645\sigma$$

for 10% defectives = 1.28
for 5% defectives = 1.645
for 2.5% defectives = 1.96
for 1% defectives = 2.33

Where,

$f_{cu,0}$: the target mean strength

$f_{cu,k}$: the specified characteristic strength

σ : the standard deviation

Concrete strength	C10 ~ C20	C25 ~ C40	C45 ~ C60
σ	4.0	5.0	6.0

4.5 Mix Proportioning

Example: (absolute volume method)

Water to Cement Ratio.

For an environment with moist freezing and thawing, the maximum water to cementitious material ratio should be 0.45. The recommended water to cementitious material ratio for the required compressive strength of 43.5 MPa is 0.31 interpolated from Table below. Since the lower water to cement ratio governs, the mix must be designed for 0.31.

Compressive strength at 28 days, MPa	Water-cementitious materials ratio by mass	
	Non-air-entrained concrete	Air-entrained concrete
45	0.38	0.30
40	0.42	0.34
35	0.47	0.39
30	0.54	0.45
25	0.61	0.52
20	0.69	0.60
15	0.79	0.70

4.5 Mix Proportioning

Example: (absolute volume method)

Air Content. For a severe freeze-thaw exposure, Table recommends a target air content of 6.0% for a 25-mm aggregate. Therefore, design the mix for 5% to 8% air and use 8% (or the maximum allowable) for batch proportions.

Target Air Content Requirements for Different Nominal Maximum Sizes of Aggregate

	9.5 mm	12.5 mm	19 mm	25 mm	37.5 mm	50 mm**	75 mm**
Recommended average total air content, percent, for level of exposure:†							
Mild exposure	4.5	4.0	3.5	3.0	2.5	2.0	1.5
Moderate exposure	6.0	5.5	5.0	4.5	4.5	4.0	3.5
Severe exposure	7.5	7.0	6.0	6.0	5.5	5.0	4.5

4.5 Mix Proportioning

Example: (absolute volume method)

Slump. The slump is specified at 25 mm to 75 mm. Use 75 mm ±20 mm for proportioning purposes.

Water Content. Table recommend that a 75-mm slump, air-entrained concrete made with 25-mm nominal maximum-size aggregate should have a water content of about 175 kg/m^3. However, rounded gravel should reduce the water content of the table value by about 25 kg/m^3. Therefore, the water content can be estimated to be about 150 kg/m^3. In addition, the water reducer will reduce water demand by 10% resulting in an estimated water demand of 135 kg/m^3.

Approximate Mixing Water Requirements for Different Slumps and Nominal Maximum Sizes of Aggregate

Slump, mm	Water, kilograms per cubic meter of concrete, for indicated sizes of aggregate*						
	9.5 mm	12.5 mm	19 mm	25 mm	37.5 mm	50 mm**	75 mm**
	Air-entrained concrete						
25 to 50	181	175	168	160	150	142	122
75 to 100	202	193	184	175	165	157	133
150 to 175	216	205	197	184	174	166	154

4.5 Mix Proportioning

Example: (absolute volume method)

Cement Content. The cement content is based on the maximum water-cement ratio and the water content. Therefore, 135 kg/m^3 of water divided by a water-cement ratio of 0.31 requires a cement content of 435 kg/m3; this is greater than the 335 kg/m^3 required for frost resistance.

❑ For severe freeze-thaw, deicer, and sulfate exposures, it is desirable to specify: (1) a minimum cementing materials content of 335 kg per cubic meter of concrete, and (2) only enough mixing water to achieve the desired consistency without exceeding the maximum water-cementing materials ratios.

4.5 Mix Proportioning

Example: (absolute volume method)

Coarse-Aggregate Content. The quantity of 25-mm nominal maximum-size coarse aggregate can be estimated from Table. The bulk volume of coarse aggregate recommended when using sand with a fineness modulus of 2.80 is 0.67. Since it has a bulk density of 1600 kg/m^3, the ovendry mass of coarse aggregate for a cubic meter of concrete is:

$$1600 \times 0.67 = 1072 \text{ kg}$$

Nominal maximum size of aggregate, mm (in.)	Bulk volume of dry-rodded coarse aggregate per unit volume of concrete for different fineness moduli of fine aggregate*			
	2.40	2.60	2.80	3.00
9.5 (⅜)	0.50	0.48	0.46	0.44
12.5 (½)	0.59	0.57	0.55	0.53
19 (¾)	0.66	0.64	0.62	0.60
25 (1)	0.71	0.69	0.67	0.65
37.5 (1½)	0.75	0.73	0.71	0.69
50 (2)	0.78	0.76	0.74	0.72
75 (3)	0.82	0.80	0.78	0.76
150 (6)	0.87	0.85	0.83	0.81



4.5 Mix Proportioning

Example: (absolute volume method)

Admixture Content. For an 8% air content, the air entraining admixture manufacturer recommends a dosage rate of 0.5 g per kg of cement. From this information, the amount of air-entraining admixture per cubic meter of concrete is:

$$0.5 \times 435 = 218 \text{ g or } 0.218 \text{ kg}$$

The water reducer dosage rate of 3 g per kg of cement results in

3 x 435 = 1305 g or 1.305 kg of water reducer per cubic meter of concrete

4.5 Mix Proportioning

Example: (absolute volume method)

Fine-Aggregate Content.

Water	=	$\frac{135}{1\times1000} = 0.135m^3$
Cement	=	$\frac{435}{3.0\times1000} = 0.145m^3$
air	=	$\frac{8.0}{100} = 0.08m^3$
Coarse aggregate	=	$\frac{1072}{2.68\times1000} = 0.4m^3$

Total volume of known ingredients: $0.76m^3$

4.5 Mix Proportioning

Example: (absolute volume method)

The calculated absolute volume of fine aggregate is then: 1 – 0.76 = 0.24 m^3, the mass of dry fine aggregate is: 0.24 x 2.64 x 1000 = 634 kg
The mixture then has the following proportions before trial mixing for one cubic meter of concrete:

Water	=	135kg
Cement	=	435kg
Coarse aggregate (dry)	=	1072kg
fine aggregate (dry)	=	634kg
Total mass	=	2276kg
Air entraining admixture	=	0.218kg
Water reducer	=	1.305kg

Estimated concrete density (using SSD aggregate) = 135+435+(1072x1.005)+(634x1.007) =2286kg/m^3

4.5 Mix Proportioning

Example: (absolute volume method)

Moisture. Corrections are needed to compensate for moisture in and on the aggregates. In practice, aggregates will contain some measurable amount of moisture. The mixing water added to the batch must be reduced by the amount of free moisture contributed by the aggregates.

With the aggregate moisture contents (MC) indicated, the trial batch aggregate proportions become:

Coarse aggregate (2% MC) = 1072 x 1.02 = 1093 kg
Fine aggregate (6% MC) = 634 x 1.06 = 672 kg

4.5 Mix Proportioning

Example: (absolute volume method)

Water absorbed by the aggregates must be excluded from the water adjustment. Surface moisture contributed by the coarse aggregate amounts to 2% – 0.5% = 1.5%; that contributed by the fine aggregate is, 6% – 0.7% = 5.3%. The estimated requirement for added water becomes:

$$135 - (1072 \times 0.015) - (634 \times 0.053) = 85 \text{ kg}$$

4.5 Mix Proportioning

Example: (absolute volume method)

The estimated batch weights for one cubic meter of concrete are revised to include aggregate moisture as follows:

Water (to be added)	=	85kg
Cement	=	435kg
Coarse aggregate (2% MC, wet)	=	1093kg
fine aggregate (6% MC, wet)	=	672kg
Total	**=**	**2285kg**
Air entraining admixture	=	0.218kg
Water reducer	=	1.305kg

4.5 Mix Proportioning

Example: (absolute volume method)

Trial Batch:

For a laboratory trial batch it is convenient to scale down the weights to produce 0.1 m^3 of concrete as follows:

Water	=	85x0.1	=	8.5kg
Cement	=	435x0.1	=	43.5kg
Coarse aggregate (wet)	=	1093x0.1	=	109.3kg
fine aggregate (wet)	=	672x0.1	=	67.2kg
Total mass			=	228.5kg
Air entraining admixture	=	0.218x0.1	=	21.8g
Water reducer	=	1.305x0.1	=	130g

4.5 Mix Proportioning

Example: (absolute volume method)

The above concrete, when mixed, had a measured slump of 100 mm, an air content of 9%, and a density of 2274 kg per cubic meter. During mixing, some of the premeasured water may remain unused or additional water may be added to approach the required slump. In this example, although 8.5 kg of water was calculated to be added, the trial batch actually used only 8.0 kg. The mixture excluding admixtures therefore becomes

Water	=	8kg
Cement	=	43.5kg
Coarse aggregate (2% MC)	=	109.3kg
fine aggregate (6% MC)	=	67.2kg
Total mass	**=**	**228kg**

4.5 Mix Proportioning

Example: (absolute volume method)

The mixing water content is determined from the added water plus the free water on the aggregates and is calculated as follows:

Water added	=	8kg
Free water on coarse aggregate $= \frac{109.3}{1.02} \times 0.015$	=	1.61kg
Free water on fine aggregate $= \frac{67.2}{1.06} \times 0.053$	=	3.36kg
Total water	**=**	**12.97kg**

The mixing water required for a cubic meter of the same slump concrete as the trial batch is: $\frac{12.97}{0.1} = 129.7kg$

4.5 Mix Proportioning

Example: (absolute volume method)

Batch Adjustments. Adjust and reestimate the amount of air-entraining admixture required for an 8% air content and adjust the water to obtain a 75-mm slump.

The adjusted mixture water for the reduced slump and air content is (3 kg water x 1 percentage point difference for air) – (2 kg water x 25/10 for slump change) + 129 = 127 kg of water.

The new cement content is:



$$\frac{127}{0.31} = 410kg$$

4.5 Mix Proportioning

Example: (absolute volume method)

The new adjusted batch weights based on the new cement and water contents are calculated after the following volume computations:

Water	=	$\frac{127}{1\times1000}$	$0.127m^3$
Cement	=	$\frac{410}{3.0\times1000}$	$0.137m^3$
Coarse agg. (dry)	=	$\frac{1072}{2.68\times1000}$	$0.4m^3$
Air	=	$\frac{8}{100}$	$0.08m^3$
Total			$0.744\ m3$

Fine aggregate volume = 1 – 0.744 = 0.256 m^3

The weight of dry fine aggregate required is 0.256 x 2.64 x 1000 = 676 kg

Air-entraining admixture (the manufacturer suggests reducing the dosage by 0.1 g to reduce air 1 percentage point) = 0.4 x 410 = 164 g or mL

Water reducer = 3.0 x 410 = 1230 g or mL

4.5 Mix Proportioning

Example: (absolute volume method)

Adjusted batch weights per cubic meter of concrete are:

Water	127 kg
Cement	410 kg
Coarse aggregate (dry)	1072 kg
Fine aggregate (dry)	676 kg
Total	2285 kg
Air-entraining admixture	164 g or mL
Water reducer	1230 g or mL
Estimated concrete density (aggregates at SSD)	= 127 + 410 + (1072 x 1.005) + (676 x 1.007) = 2295 kg/m3

After checking these adjusted proportions in a trial batch, it was found that the concrete had the desired slump, air content, and yield. The 28-day test had an average compressive strength of 48 MPa, which exceeds the required compressive strength of 43.5 MPa.

Chapter Four：Concrete

1. Introduction
2. Compositions
3. Structure
4. Strength and Damage
5. Mix Proporationing
6. **Fresh and Early-age Properties**
7. Volume Changes
8. Durability

Summary

1. Strength

Effect of air content, curing temperature, moisture condition on concrete Strength

Mixture temperature

Strength test method, size effect

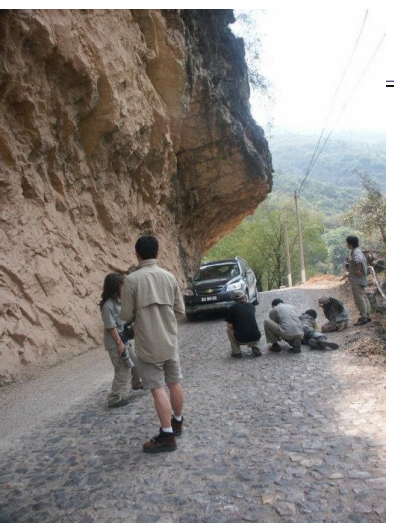
2. Proportioning

The three principal requirements

Overall procedure

W/c ratio, water demand, cement content





Questions?